

COOLSHADE · REQUIREMENTS

CoolShade requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from CSH-CAL-001 for every requirement; R6 thresholds adopted under CSH-DDR-001; R13 stated against \$700 and the proposed \$750
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

CoolShade requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be revised after co-design sessions (see CSH-PRB-001). The status column comes from the TRL 3 calculations in CSH-CAL-001 v0.2: no requirement is not met, three are at risk (R9 water hygiene, R13 cost, R16 scaling), one cannot be judged at TRL 3 (R5 wetting, for lack of droplet data) and thirteen are met, four of them by design and R10 on paper. On 2026-09-25 Amish accepted the recommendations (CSH-DDR-002), which changed three targets: R4 is restated as 2 °C while spraying (was 2 °C averaged, not met at 1 m/s); R10 is restated as surviving 30 m/s with the fabric removed, with the fabric fitted only in forecast gusts below 15 m/s (was 30 m/s with the fabric fitted, not met); and R13 is \$775 (was \$700). The R6 thresholds are decided by Amish (CSH-DDR-001, D2).

Design day: 38 °C air, 25 % relative humidity, clear sky, wind about 1 m/s, 5 h of misting demand, 4.5 peak sun hours. **Humid edge:** 32 °C and 60 % relative humidity, the wettest air in which the controller still mists.

Table 1. Requirements, targets and status after the TRL 3 calculations (CSH-CAL-001 v0.2; tags in brackets are lines of its script output).

ID	Requirement	Target	Verification (TRL 3 or later)	Status (CSH-CAL-001)
R1	Shaded waiting area	6 m ² or more of shade at solar noon; room for at least 6 people standing or seated and one wheelchair space of 0.8 x 1.2 m	Plan layout; sun path check per site	Met: shadow 6.30 m ² or more at any sun elevation of 45° or more; room for 7 standing, 3 seated and a wheelchair [A2, A3]. Where the shadow falls depends on the site
R2	Headroom	Clear height 2.2 m or more under any part of the frame, nozzle or brace in the waiting area	Model check	Met: 2,234 mm under the shortened rear knee braces; nozzle tips 2,584 mm or higher [B2, B3]
R3	Radiant heat reduction	Mean radiant temperature under the canopy at least 15 °C below full sun at midday on the design day	Literature comparison; later field measurement with a globe thermometer (HeatMap Node method)	Met on published evidence: about 17.3 °C for shade sails in Tempe (Middel et al., 2021); unverified for this fabric
R4	Air cooling from misting	Air temperature in the waiting zone (0.5 to 2.0 m height) at least 2 °C below ambient while spraying on the design day (restated, CSH-DDR-002; was averaged over the spray cycle)	Energy balance at TRL 3; later field test	Met: 2.71 °C while spraying at 1 m/s, held to 1.35 m/s; 1.35 °C averaged over the cycle [E2, E3]
R5	No wetting	No perceptible wetting of skin or clothing at 1.5 m height on the design day; no puddles on the waiting area	Droplet size data from nozzle maker; later field test	Not verifiable at TRL 3: met only if the nozzle's Dv0.9 is below about 81 µm at 7 bar; no droplet data yet [D3]
R6	Smart activation	Mists only when air temperature is 32 °C or more, relative humidity is 60 % or less, presence is detected in the last 2 min and the tank is above its low level; stops within 60 s when any condition fails	Control logic review; later bench test	Met by design: stops within 30 s [I1]; thresholds decided (CSH-DDR-001, D2)
R7	Water use	100 L or less per design day	Nozzle flow calculation	Met: 85 L used, 95 L drawn with the drain-and-refill routine [C2, C5]; holds up to 4.75 L/h per nozzle [C3]
R8	Energy autonomy	Energy-neutral on the design day from the onboard panel; battery covers one full design day with no sun	Energy budget	Met: 203.8 Wh used against 337.5 Wh harvested; 25 Ah battery (CSH-DDR-002; was 20 Ah, 1.005 days) covers 1.256 design days; no charging in air above about 41 °C [F2, F5]
R9	Water hygiene	Potable-quality supply water; 5 µm filtration; mist line drained within 5 min after every session; water in the tank 48 h old or less; opaque, shaded, lockable tank; documented weekly disinfection	Design review; later microbiological sampling with a partner lab	At risk: line drains in about 27 s with the vacuum breaker and water stays under 24 h old with the routine [C5, C7], but tank water sits at about 34 °C, in the

ID	Requirement	Target	Verification (TRL 3 or later)	Status (CSH-CAL-001)
				20 to 50 °C <i>Legionella</i> growth range (WHO) [I3]
R10	Wind resistance	Frame and anchors survive a 3 s gust of 30 m/s (108 km/h) with the fabric removed; fabric fitted only when forecast gusts are below 15 m/s (54 km/h); fabric removable in 15 min or less (restated, CSH-DDR-002; was 30 m/s with the fabric fitted)	Structural calculation at TRL 3	Met on paper: fabric off at 30 m/s, posts 0.14 and purlins 0.15 utilization [G13]; fabric on at 15 m/s, long beams 0.88 [G14]; anchors need about 2.8 kN each, to be confirmed per site. The original case gives 3.52 on the beams [G7]
R11	Privacy	No camera or microphone; presence by passive infrared only; if a status link is fitted, only counts and levels (on-time, water used, temperature, humidity, faults) leave the device	Design review	Met by design; no status link fitted (CSH-DDR-001, D7)
R12	Electrical safety	12 V DC only, no mains; battery fused at the terminal; all electronics in a lockable IP65 enclosure; LiFePO4 chemistry with BMS and cold-charge cutoff	Design review	Met by design: 15 A terminal fuse, margin 2.0 on the largest current [I2]
R13	Cost	Parts for one prototype \$775 or less (<code>budget_usd</code> , set by Amish under CSH-DDR-002; was \$700)	BOM	At risk: \$773, \$2 (0.3 %) under; steel prices look low [J1, J2]
R14	Buildability	Bolted assembly; no welding; built with a drill, angle grinder, spanners and a ladder by two people in 2 days or less	Assembly review	Met by design: about 48 bolts and anchors, no welds [H2]; build time unverified
R15	Refill interval	1 design day or more between tank refills	Water balance	Met: 1.41 design days per full tank [C4]
R16	Maintenance	Nozzles cleanable or replaceable without tools in 15 min or less; filter changed in 10 min or less; descaling interval 1 month or longer	Design review; later field log	At risk in hard-water areas: scale can block 0.4 mm orifices; not verifiable at TRL 3
R17	Movability	Can be unbolted and moved to another site by two people in 1 day or less; no part heavier than 40 kg to lift (tank empty)	Mass estimate	Met: heaviest part a front post with brace, 22.0 kg [H1]

■ Assumptions

- Nozzle flow 4 L/h each at 7 bar, eight nozzles, 50 % spray duty cycle while active; this implies a discharge coefficient of 0.24 for a 0.4 mm orifice, to be checked against nozzle datasheets.
- About 87 % of the sprayed water evaporates in the air; the rest drifts away, drips or wets surfaces (estimate).
- Cooling is estimated over a control volume 3.0 m long and 2.0 m high with air crossing it at the wind speed. Real mixing is less even than this.

- Panel output 100 W x 4.5 peak sun hours x 0.75 system factor; battery 12.8 V 25 Ah, 80 % usable.
 - Wind: fabric treated as solid with 5 % sag, net normal force coefficient 1.2, S275 steel with a 1.5 factor on wind.
- All other assumptions are listed in CSH-CAL-001, Table 1.